

The Pioneers or Astrophysics Pioneers Program is a NASA program started in 2020 intended to use low-cost, small-size hardware to carry out sub-orbital and space-based astrophysics investigations. The missions are expected to include SmallSats, balloon payloads, and payloads attached to the ISS, with a \$20M cost cap.

Four concepts were chosen in January 2021:

Aspera, a SmallSat to study galaxy evolution

Pandora, a SmallSat launched on January 11th, 2026, to study at least 20 stars and their respective exoplanets in order to determine their atmospherical composition

StarBurst, a SmallSat to detect high-energy gamma rays from neutron star mergers, targeted to launch in summer 2027

PUEO (Payload for Ultrahigh Energy Observations), a balloon mission successfully launched on December 19, 2025 from the McMurdo Station's Long Duration Balloon Facility in Antarctica for a 23-day flight in order to detect signals from neutrinos; future flights are possible

Three additional missions have been selected as of June 2026:

TIGERISS (Trans-Iron Galactic Element Recorder on the International Space Station), selected in 2022 and anticipated to launch in 2027, a payload attached to the ISS to test models of nucleosynthesis and increase understanding of the production and distribution of elements in the galaxy

Landolt space mission, selected in 2024, a SmallSat intended to act as an artificial "star" in the Earth's orbit, enabling the absolute flux calibration of telescopes and thus more accurate measurements of brightness for astronomical objects

POEMM (Planetary Origins and Evolution Multispectral Monochromator), selected in July 2024 and expected to launch around 2028, a balloon mission with a 1.8 meter telescope and multiple spectrometers that will study planet formation and evolution in the far infrared

== References ==